



ResiliLytics Dashboard

Sourcing Intelligence for Resilient Supply Chains

ResiliLytics User Manual: SME-Native Procurement Resilience Framework

Minimal-Input, Multidimensional Risk-to-Action Architecture

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Public Research Edition

ResiliLytics User Manual: SME-Native Supply Chain Resilience Intelligence Framework

1. Introduction — Why Resilience Must Be Structured

Supply chain disruption is no longer an exceptional event.

For Small and Medium Enterprises (SMEs), disruption is not merely operational inconvenience, it is structural vulnerability. Where large enterprises may absorb volatility through scale and diversification, SMEs often operate within tighter supplier networks, narrower geographic footprints, and thinner financial margins.

ResiliLytics was designed in response to that reality.

It was built on a simple but powerful premise:

- Resilience should not require enterprise-level infrastructure.
- From minimal supplier data, meaningful structural intelligence can be derived.
- ResiliLytics transforms basic sourcing inputs into a structured resilience posture assessment and, critically, into sequenced mitigation guidance.
- This is not a visualization tool alone.

It is a decision architecture, applying SCOR-DS Source process principles to deliver what large enterprises pay thousands for, free to every SME.

2. Origin & Research Context

ResiliLytics was developed as an original research contribution focused specifically on SME supply chain resilience.

Existing resilience analytics systems frequently depend on:

- ERP integration
- Complex multi-system data ingestion
- Advanced enterprise analytics infrastructure

SMEs often lack these capabilities.

ResiliLytics demonstrates an alternative model aligned with SCOR-DS Source process principles:

Minimal Input

→ Multidimensional Structural Inference

→ SME-Calibrated Weighted Architecture

- Constraint Classification
- Risk-to-Action Sequencing
- Controlled Analytical Expansion

The framework was informed by documented SME disruption patterns, including:

- Supplier concentration fragility
- Geographic clustering exposure
- Volatility amplification effects
- Limited sourcing optionality
- Margin transmission sensitivity

Its contribution lies in the integrated architecture that connects structural exposure to sequenced action, and in making that architecture accessible to every SME, free of charge.

3. The Four-Layer Architecture

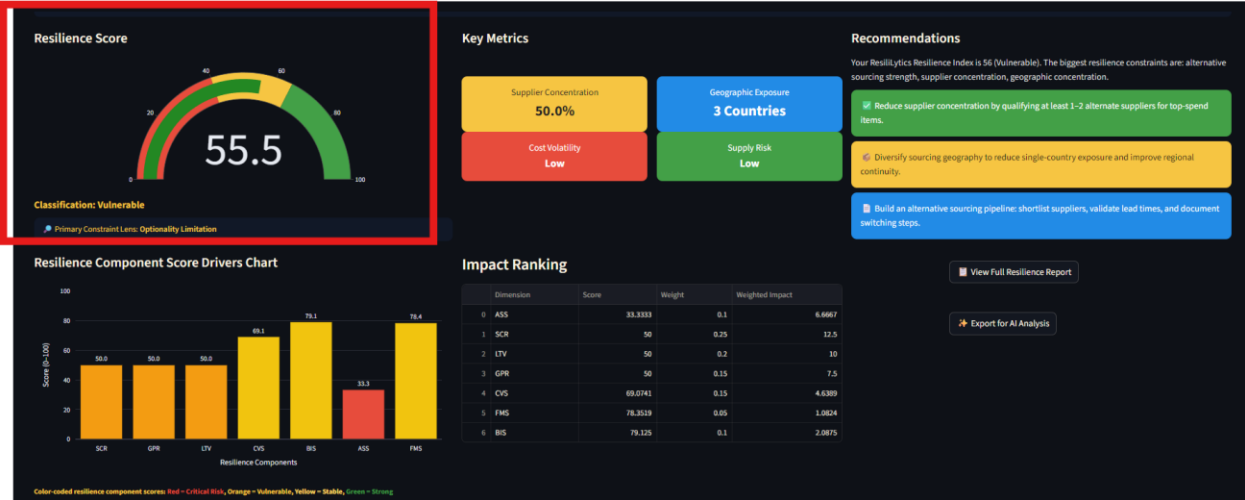
ResiliLytics operates across four integrated layers.

Layer 1 — Monitoring

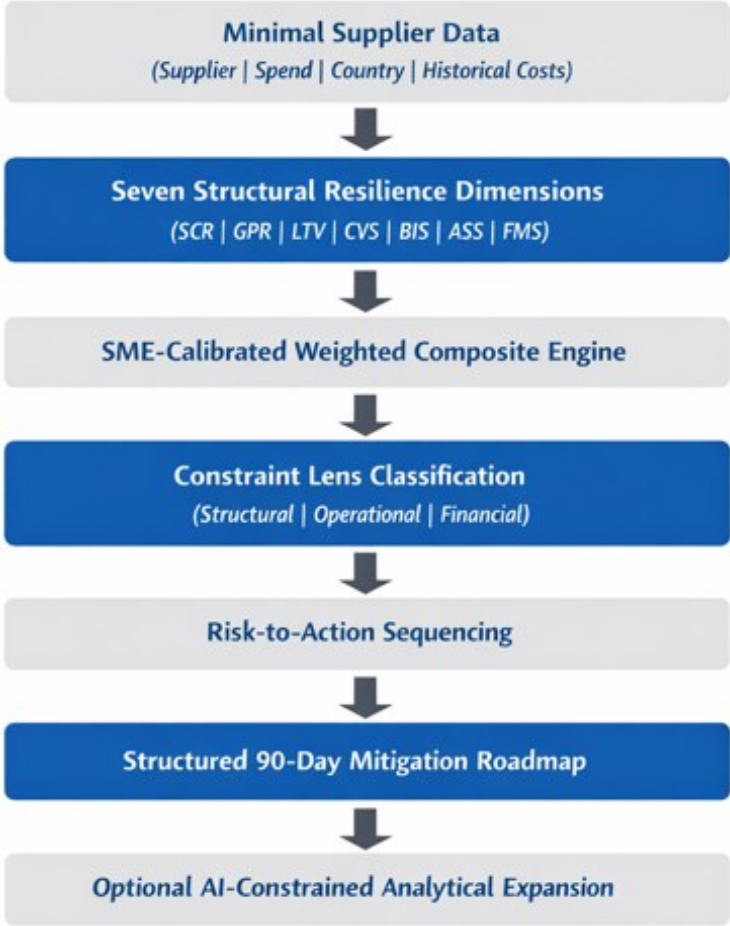
The system calculates:

- A Composite Resilience Score (0–100)
- Key Metrics
- Resilience component score drivers chart
- Seven structural resilience dimensions
- A weighted impact prioritization table

This provides immediate structural visibility.



Conceptual ResiliLytics Risk-to-Action Architecture



Layer 2 — Diagnosis

Beyond visibility, the framework identifies:

- The weakest resilience dimensions
- The dominant systemic constraint lens
- The likely risk transmission pathway
- Recommendations

This transforms observation into structural understanding.

Executive Insight

Executive Summary: Composite Score 56 (Vulnerable). Elevated structural risk exposure. Primary constraint lens: **Structural resilience**. Key limiting dimensions: Alternative Sourcing Strength, Supplier Concentration Resilience, Lead-Time Stability.

Diagnosis: • **SCR — Supplier Concentration Resilience**: High concentration reduces continuity if a single supplier fails or reprioritizes capacity. • **LTV — Lead-Time Stability**: Timing variability reduces planning reliability and increases reactive expediting and schedule churn. • **GPR — Geographic Portfolio Resilience**: Single-country exposure increases disruption sensitivity (logistics, policy, FX, and regional shocks).

Advanced Diagnostics

- Structural Resilience Score: 44.4
- Operational Stability Score: 64.6
- Financial Exposure Score: 73.7

Risk Transmission Pathway

- **Identified Weakness**: ASS
 - **Likely Consequence**: Slow supplier switching during disruption events.
 - **Intervention Direction**: Document and validate supplier-switch playbooks.
-

Layer 3 — Prescriptive Automation

ResiliLytics automatically generates:

- Three prioritized mitigation actions
- A structured 90-day roadmap
- Sequenced intervention logic aligned with the dominant constraint

This is where the system moves from assessment to structured action.

Priority Mitigation Actions

1. Reduce top-supplier dependency: qualify 1–2 alternates for the highest-spend inputs and shift a controlled share of volume.
2. Improve operational stability: introduce disruption triggers, reliability scoring, and escalation rules for late/variable suppliers.
3. Reduce single-country exposure: introduce at least one alternate sourcing geography for the most critical categories.

90-Day Resilience Roadmap

- 0–30 Days: Confirm top concentration exposures (supplier + country); shortlist alternates for critical categories.
- 30–60 Days: Validate alternates (lead time, capacity, compliance); shift a small controlled volume to reduce structural dependency.
- 60–90 Days: Finalize supplier-switch playbook, add regional redundancy for key inputs, and set a quarterly resilience review cadence.

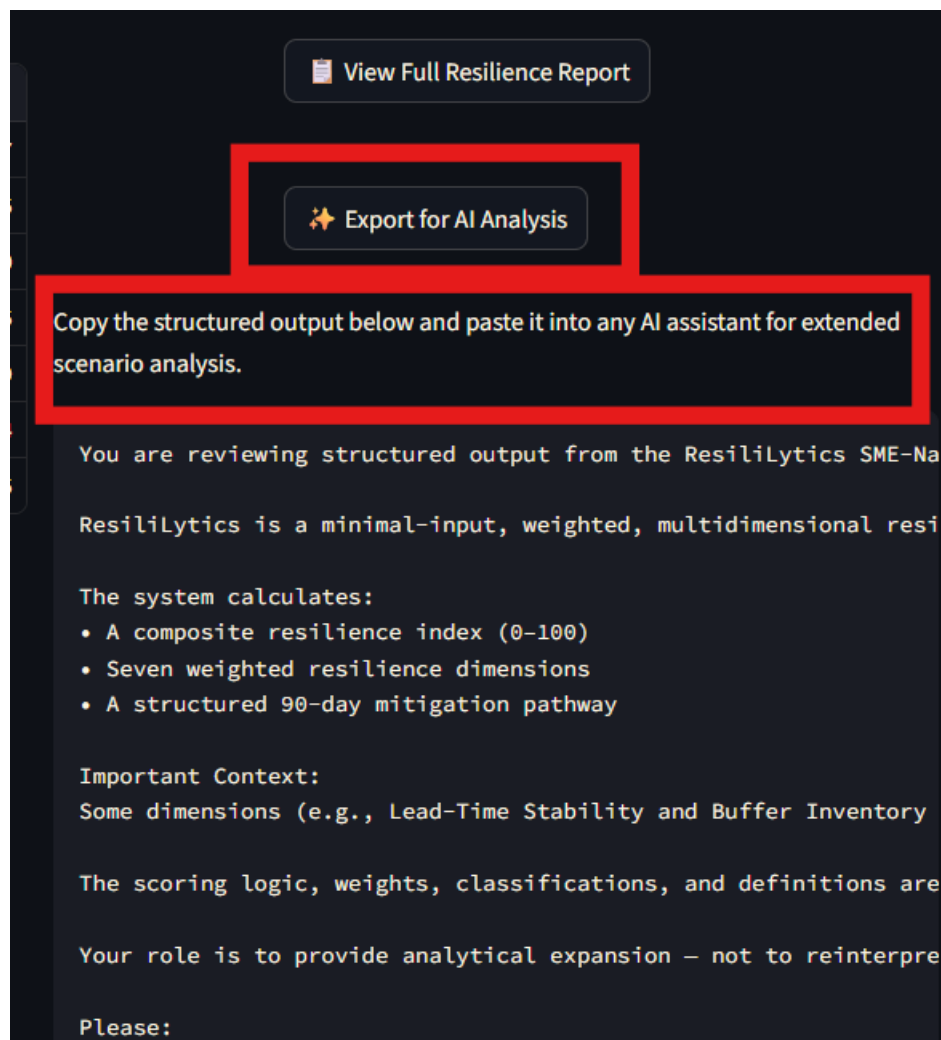
Layer 4 — Optional AI-Assisted Expansion

For deeper strategic interpretation, users may copy a structured analytical prompt into an AI tool like ChatGPT, Google Gemini, Claude etc.

Important safeguards:

- No uploaded data is transmitted externally

- Only structured summary metrics are manually copied
- AI output does not override the model



4. Data Requirements

ResiliLytics intentionally minimizes data burden.

Required columns:

Column	Description
Supplier	Supplier name
Spend	Total spend allocation
Country	Supplier location
Historical Costs	Semi-colon separated historical cost values

Example format:

12.5; 12.8; 13.1; 12.9

Historical cost data strengthens volatility-sensitive dimensions but is not mandatory.

All processing occurs within the active browser session. No uploaded data is stored.

5. The Seven Resilience Dimensions

Each dimension is normalized on a 0–100 scale.

Higher values represent stronger structural resilience.

I. Supplier Concentration Resilience (SCR)

Measures dependency on dominant suppliers.

II. Geographic Portfolio Resilience (GPR)

Captures exposure to single-country or regionally concentrated sourcing. Measures both how concentrated spend is geographically and how diversified the sourcing geography is across countries. Higher geographic diversification across a broader range of countries produces higher GPR scores. Single-country concentration (regardless of which country) significantly reduces this score.

III. Lead-Time Stability Proxy (LTV)

Interaction-derived structural indicator reflecting timing instability risk under structural stress conditions.

IV. Cost Volatility Sensitivity (CVS)

Measures exposure to historical price variability.

V. Buffer Inventory Strength (BIS)

Interaction-derived structural indicator of shock absorption capacity under volatility and concentration pressure.

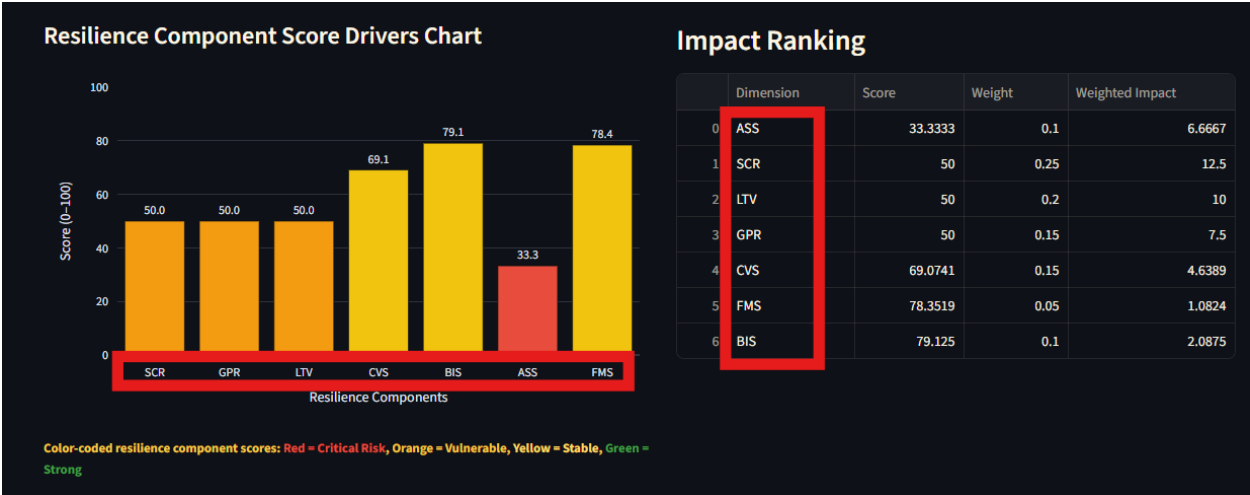
VI. Alternative Sourcing Strength (ASS)

Reflects supplier optionality and switching flexibility.

VII. Financial Margin Sensitivity (FMS)

Interaction-derived structural indicator of how volatility and concentration transmit into profitability pressure under disruption conditions .

These indicators are derived structural measures inferred from concentration and variability relationships within the dataset.



6. Weighted Composite Framework

The Composite Resilience Index integrates all seven dimensions using SME-calibrated weights:

- SCR — 25%
- GPR — 15%
- LTV — 20%
- CVS — 15%
- BIS — 10%
- ASS — 10%
- FMS — 5%

These weights reflect empirically observed SME disruption dynamics in which concentration and timing instability often precede financial strain.

7. Adaptive Concentration Mode

When supplier concentration exceeds a structural fragility threshold (50%), the system activates adaptive weighting.

In this mode:

- Concentration emphasis increases
- Financial sensitivity emphasis increases
- Remaining weights adjust proportionally

This prevents linear averaging from understating high-dependency exposure.

Specific recalibration mechanics remain internal to the framework.

8. Understanding Your Score

Score Range	Classification
0–40	Critical Risk
41–65	Vulnerable
66–85	Stable
86–100	Resilient

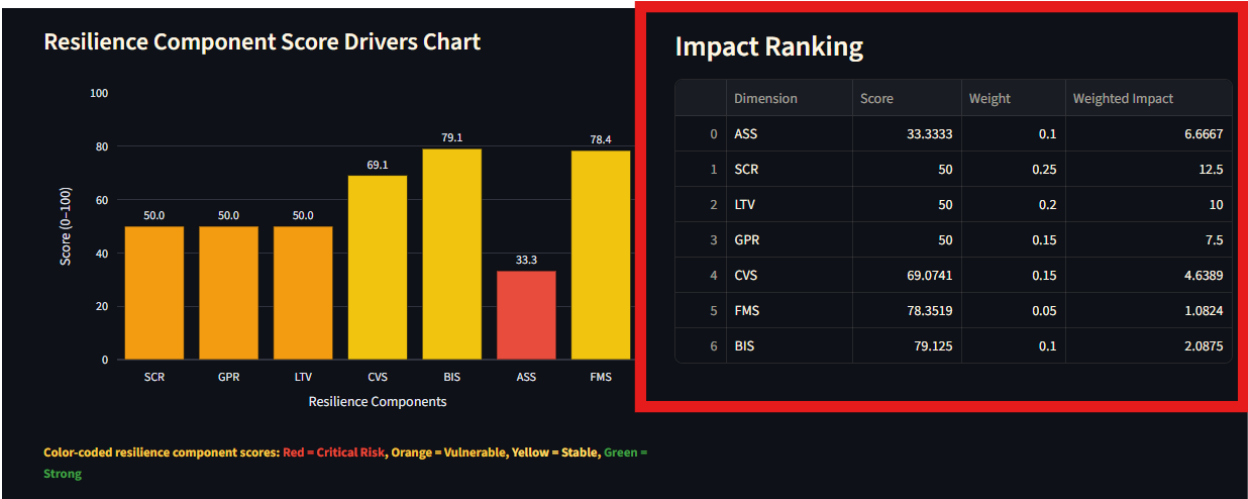
The score reflects structural posture — not guaranteed future performance.

9. Impact Ranking

The Impact Ranking table displays:

- Raw dimension score
- Active weight
- Weighted impact contribution

This allows mitigation to be prioritized logically rather than reactively.



10. Risk-to-Action Sequencing Architecture

ResiliLytics is designed around a structured Risk-to-Action principle.

Identified weaknesses trigger:

1. Constraint classification
2. Consequence mapping
3. Prioritized mitigation sequencing
4. Structured 90-day action planning

This ensures that risk visibility translates into decision-ready intervention.

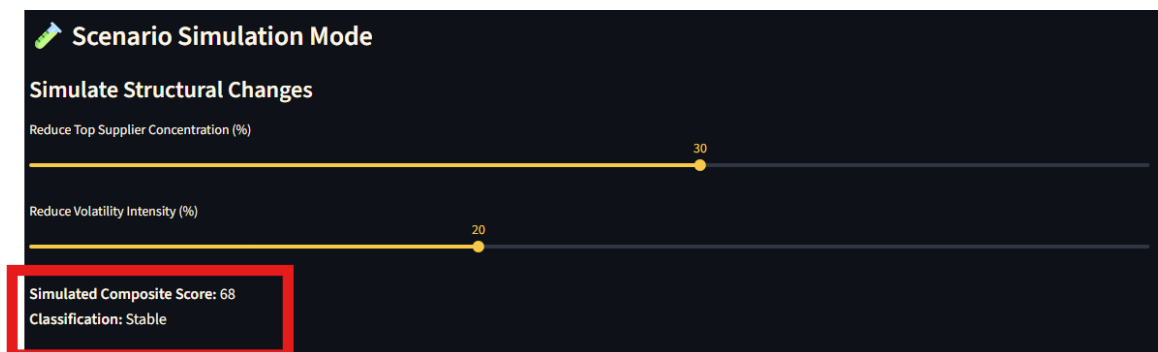
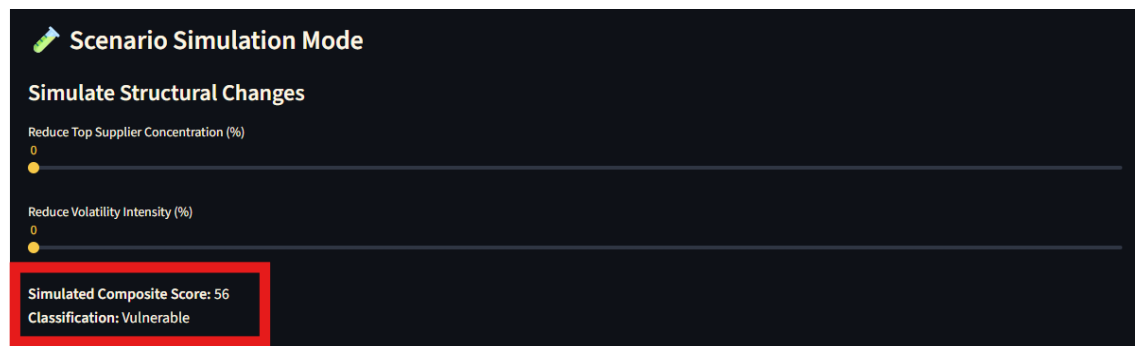
11. Scenario Simulation Mode

Users may simulate:

- Reduced supplier concentration
- Reduced volatility intensity

The system recalculates structural posture instantly.

This enables proactive strategic testing before implementation.



12. Model Assumptions & Boundaries

ResiliLytics:

- Measures structural procurement exposure — not operational performance
- Does not forecast revenue or predict insolvency
- Does not predict disruption events or guarantee supply continuity
- Does not claim full SCOR-DS compliance — it applies SCOR-DS Source process principles

Four dimensions (LTV, BIS, ASS, FMS) are interaction-derived structural indicators, inferred from the mathematical relationship between concentration and volatility signals within the uploaded dataset. This is an explicitly acknowledged and documented modeling approach, not a limitation to be hidden.

The framework is calibrated specifically for SME environments and is the only free, firm-level procurement resilience tool of its kind currently available.

13. Data Privacy & Security

- No uploaded data is stored
- No session data is retained
- No automatic external transmission occurs
- AI expansion requires manual metric copying

ResiliLytics is a free, non-commercial research-validated framework. No data is stored, monetized, or shared

14. Intended Use & Contribution

ResiliLytics demonstrates that meaningful resilience intelligence can be derived from minimal structured input.

Its contribution lies in system architecture:

Minimal Input

- Multidimensional Structural Inference
- Adaptive Weighting
- Constraint Classification

- Risk-to-Action Sequencing
- Controlled Generative Expansion

It is intended for strategic procurement evaluation, resilience planning, and research contribution, accessible to every SME, free of charge, without specialist infrastructure. ResiliLytics applies SCOR-DS Source process principles, making enterprise-grade procurement resilience intelligence available to the businesses that need it most